

[CONFIDENTIAL] MISSION BRIEFING

1. SITUATION REPORT (SITREP)

Incident Summary: At 04:00 hours, UMD network sensors detected unauthorized encrypted traffic leaving a dormant Google Cloud project. Intelligence suggests a rogue actor, codenamed "**Iron Shell**," attempted to establish a persistent multi-cloud bridge between a Google Cloud frontend and an AWS backend database to exfiltrate sensitive research data.

Current Status: The adversarial infrastructure has been partially neutralized. The rogue instances were terminated, but the **data payload (The Flag)** remains locked inside a snapshot of their database configuration.

The Problem: Iron Shell utilized a "Broken Bridge" architecture. The frontend and backend are isolated across different cloud providers with strict network segmentation. To recover the flag, we must recreate their environment exactly as it was, establish the network link, and extract the payload.

2. MISSION OBJECTIVES

You are tasked with Forensic Reconstruction. You must rebuild the adversary's infrastructure to recover the flag.

PHASE I > DIGITAL FORENSICS (Reconnaissance)

Intel indicates the adversary stashed their deployment artifacts (Source Code & Infrastructure-as-Code templates) in a **Google Cloud Storage Bucket** within this project to avoid detection.

- **Target:** Locate the bucket. Naming convention is known to contain: **instructions-multi-cloud-gcp-aws-umd**.
- **Action:** Authenticate via CLI and exfiltrate (download) all assets to your local workstation.

PHASE II > BACKEND RECONSTRUCTION (AWS)

The adversary stored the data payload in an AWS RDS PostgreSQL instance.

- **Action:** Use the recovered CloudFormation Template (`multi-cloud-backend.yaml`) to provision the backend vault in `us-east-1`.
- **Security Constraint:** You must set a Master Password. Do not lose this.
- **Intel Note:** The adversary used an automated "Seeding" mechanism. The database will appear empty until the frontend successfully creates a connection.

PHASE III > FRONTEND DEPLOYMENT (GCP)

We need a "Jumpbox" to talk to the database.

- **Action:** Provision a standard Compute Engine instance (`e2-micro`) in the `us-east1` region.
- **Payload Delivery:** Deploy the recovered Python Source Code onto this instance.
- **Configuration:** The application requires environment variables to locate the AWS target. Create a `.env` file (refer to the `.env.example` found in the artifacts).

PHASE IV > BRIDGE ENGINEERING (Networking)

[CRITICAL FAILURE POINT] The adversary intentionally severed the network link. The GCP Frontend cannot currently talk to the AWS Backend.

- **Task:** The Web Dashboard runs on `Port 5000`. Configure GCP firewalls to allow your personal laptop to access the dashboard.
- **The Fix:** The application will report a "Connection Timeout" (Sad Face). You must analyze the AWS Security Group created by the CloudFormation stack and `authorize` the GCP Instance's Public IP to cross the bridge.

PHASE V > EXFILTRATION

- **Action:** Once the bridge is established, trigger the retrieval mechanism via the Web Dashboard.
- **Success Criteria:** Decrypt and submit the Flag: `ENPM665{...}`.

3. RULES OF ENGAGEMENT (ROE)

1. **Cost Discipline:** This is a live fire exercise. To prevent budget overruns, you must **TERMINATE** all resources (Delete GCP VM, Delete AWS CloudFormation Stack) immediately upon flag capture.

PROJECT: TERP SHIELD // OPERATION BROKEN BRIDGE
CLASSIFICATION: ENPM665 EYES ONLY
DATE: 03 DEC 2025
ORIGIN: UMD CYBER COMMAND (UCC) - COLLEGE PARK HQ

2. **OpSec:** Do not leave the Web Dashboard (Port 5000) open to the entire internet (0.0.0.0/0). Whitelist only your IP.
3. **Tools:** You are authorized to use Google Cloud Shell, local terminals, and the AWS Console.

GOOD LUCK, ENGINEER. FEAR THE TURTLE. 🐢